Assignment-4

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⇒ Section-A

Q1. (i) A statistical hypothesis is an assumption or statement about population parameter that can be tested using statistical methods.

(ii) Null hypothesis : A statement that assumes no effect or no difference exists in the population.

Alternative hypothesis - A statement that contradicts the null hypothesis, suggesting an effect or difference exists.

(iii) One Tailed Test : A hypothesis test where critical region is in one tail of distribution testing for specific direction.

Two tailed test : A hypothesis test where critical region are in both tails of distribution testing for any difference.

(iv) Type I error (α) - Rejecting null hypothesis when it is actually true.

Type 2 error (β) - Failing to reject null hypothesis when it is actually false.

(v) Significance level (α) - The probablity of making a Type I error, typically set 0.05 or 0.01.

critical region - The range of values that leads to rejecting null hypothesis; it is determined by significance level.

(vi) Confidence interval - A range of values derived from sample that is likely to contain true population parameter with specified confidence level.

Confidence limits - The upper and lower bounds of the confidence interval.

Q2. Statiscal hypothesis under test is called Null Hypothesis.

Q3. A wrong decision about null hypothesis leads to Type I and type 2 errors.

Q4. The probablity of type I error is given by α (significance level).

Q5. The level of a significance of test is related to type I error and is given by α.

Q6. critical region provides us criterion for rejection of null hypothesis.

Q7. Accepting an incorrect hypothesis

Q8. Accepts a null hypothesis when it is false.

Q9. Rejects it, when it is true.

⇒ Section-B

Q10. (A) Null : $H_0 : \mu = 162.5$ cm

Alternative : $H_1 : \mu < 162.9$ cm.

(B) Null : $H_0 : \mu = 8$ min

Alternative : $H_1 : \mu \neq 8$ min.

Q11. (A) Null : The standard derivative of hotel room rates are equal ($\sigma_1^2 = \sigma_2^2$)

Alternative ⇒ $\sigma_1^2 \neq \sigma_2^2$

(B) Null : The main heat produced by two wires is the same ($\mu_1 = \mu_2$)

Alternative ⇒ $\mu_1 \neq \mu_2$.

Q12. (A) Null : $\mu = 1000$ charges

Alternative : $\mu > 1000$ charges.

(B) Null : $\mu = 50$ kg

Alternative : $\mu \neq 50$ kg.

⇒ Section-C

Q13. freq func = $f(x,\theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$ $H_0 : \theta = 1$, $H_1 : \theta = 2$

(i) $\alpha = P(x \geq 0.5 \mid \theta = 1) = \int_{0.5}^{1} f(x,1)\,dx = \int_{0.5}^{1} 1\,dx$

$$\alpha = \int_{0.5}^{1} dx = [x]_{0.5}^{1} = \underline{0.5}.$$

(ii) $\alpha = P(1 \leq X \leq 0.5 \mid \theta = 1) = \int_{1}^{1.5} f(x,1)\,dx = \underline{0.}$

→ Type 2 error (β) : probablity of failing to reject Ho when H_1 is twice.

(i) $\beta_i = P(x < 0.5 \mid \theta = 2) = \int_0^{0.5} f(x,2) = \int_0^{0.5} \frac{1}{2} dx$

$= \frac{1}{2}[x]_0^{0.5} = \frac{1}{2}[0.5] = \underline{\underline{0.25}}.$

(ii) $\beta_{ii} = P(1 \geq x \text{ or } 1.5 \leq x \mid \theta = 2) = P(x < 1 \mid \theta = 2) + P(1.5 < x \mid \theta = 2)$

$= \int_0^1 f(x,2)\,dx + \int_{1.5}^2 f(x,2)\,dx$

$= \int_0^1 \frac{1}{2} dx + \int_{1.5}^2 \frac{1}{2} dx = \frac{1}{2}(1 + 0.5)$

$= \frac{1.5}{2} = \underline{\underline{0.75}}.$

<u>Q14.</u> $H_0 : P = \frac{1}{2}$; $H_1 = p = \frac{3}{4}$; $n = 5$

$\alpha = P(\text{more than 3 heads} \mid H_0 : P = \frac{1}{2})$

$\alpha = P(X = 4 \mid P = \frac{1}{2}) + P(X = 5 \mid P = \frac{1}{2})$

$P(X = 4) = {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 = \frac{5}{2^5} = \frac{5}{32}$

$$P(x=5) = {}^5C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^0 = \frac{1}{2^5} = \frac{1}{32}$$

$$\Rightarrow \alpha = \frac{1}{32} + \frac{1}{32} = \frac{5+1}{32} = \underline{\underline{\frac{3}{16}}}.$$

$$\beta = P\left(x \leq 3 \mid p = \frac{3}{4}\right) = P(x \leq 1) + P(x=2) + P(x=3)$$

where $P(x=k) = {}^5C_k\left(\frac{3}{4}\right)^k\left(\frac{1}{4}\right)^{5-k}$

$$P(x=0) = {}^5C_0\left(\frac{3}{4}\right)^0\left(\frac{1}{4}\right)^5 = \frac{1}{1024}$$

$$P(x=1) = 54\left(\frac{3}{4}\right)^1\left(\frac{1}{4}\right)^4 = \frac{15}{1024}$$

$$P(x=2) = {}^5C_2\left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right)^3 = \frac{90}{1024}$$

$$P(x=3) = {}^5C_3\left(\frac{3}{4}\right)^3\left(\frac{1}{4}\right)^2 = \frac{270}{1024}$$

$$\Rightarrow \beta = \frac{1+15+90+270}{1024} = \frac{376}{1024} = \underline{\underline{\frac{47}{128}}}.$$

Q15. $f(x,\theta) = \theta e^{-\theta x} \{0 \leq x < \infty\}$ $\quad H_0 : \theta = 2$

$H_1 : \theta = 1$

Type 1 error (α) :-

$\alpha = P(X \geq 1 \mid \theta = 2)$

$= \int_1^\infty 2e^{-2x}\,dx = \cancel{2}\left[\frac{e^{-2x}}{\cancel{-2}}\right]_1^\infty = 1 - e^{-2x}\Big|_1^\infty$

$\alpha = \cancel{-e^{-\infty}} - (-e^{-2}) = e^{-2}$ or $\frac{1}{e^2}$

Type 2 error (β) :-

$\beta = \beta(X < 1 \mid \theta = 1)$

$= \int_0^1 1e^{-1x}\,dx = 1\left|\frac{e^{-x}}{-1}\right|_0^1 = 1 - e^{-x}\Big|_0^1$

$\beta = -e^{-1} - (-e^{-0}) = \boxed{1 - e^{-1}}$

$\Rightarrow \beta = 1 - \frac{1}{e} = \boxed{\frac{e-1}{e}}$.

Q16. $f(x,\theta) = \frac{1}{\theta} e^{-x/\theta} \; \{0 \leq x \leq \infty, \; \theta > 0)$

$H_0 : \theta = 2 \Rightarrow$ exponential (Rejected)

$H_1 : \theta = 4 \Rightarrow$ Accepted.

$\Rightarrow$ Acceptance Region $= \boxed{x < 6}$

$\Rightarrow$ Type I error (α) –

$$\alpha = P(X \geq 6 \mid \theta = 2) = \int_6^\infty f(x,2)\,dx = \int_6^\infty \frac{1}{2} e^{-x/2}\,dx$$

$$= \frac{1}{\cancel{2}} \left[\frac{e^{-x/2}}{-1/\cancel{2}} \right]_\infty^6 = \cancel{-e^{-\infty}} - (-e^{-6/2}) = e^{-3}.$$

$$= \boxed{\frac{1}{e^3}}.$$

$\Rightarrow$ Type 2 error (β) –

$$\beta = P(X < 6 \mid \theta = 4) = \int_0^6 f(x,4)\,dx$$

$$= \left[\frac{1}{4} e^{-x/4} \right]_0^6 = \frac{1}{\cancel{4}} \left[\frac{e^{-x/4}}{-1/\cancel{4}} \right]_0^6$$

$$\beta = -e^{-6/4} - (-e^0) = \boxed{1 - e^{-3/2}}.$$